Journal of Development Economics 105 (2013) 226–236

Contents lists available at ScienceDirect

Journal of Development Economics

journal homepage: www.elsevier.com/locate/devec

Living longer: The effect of the Mexican conditional cash transfer program on elderly mortality[☆]

Tania Barham [a,b,*], Jacob Rowberry [c]

[a] Department of Economics, University of Colorado at Boulder, USA
[b] Institute of Behavioral Science, University of Colorado at Boulder, USA
[c] Ford School of Public Policy, University of Michigan, USA

ARTICLE INFO

Article history:
Received 18 July 2012
Received in revised form 4 August 2013
Accepted 6 August 2013

JEL classification:
O15
I18
I38
J18

Keywords:
CCT
Mortality
Elderly
Health
Mexico
Progresa
Oportunidades

ABSTRACT

With both an aging population and a transition from communicable to chronic diseases, the health of the elderly is a growing issue in many developing countries. Conditional cash transfer programs are usually thought to benefit young people, but may also benefit other age groups since some programs require that all household members have regular preventive health check-ups. This paper exploits the phasing-in of the Mexican conditional cash transfer program, Progresa, between 1997 and 2000, and shows a 4% decline in average, municipality-level mortality for people aged 65 and older. The program not only reduced deaths due to more traditional infectious diseases, but also diabetes related deaths. Given that diabetes deaths are a leading cause of death in Mexico, and in the top 10 causes of death in many high- and middle-income countries, this is an important finding.

1. Introduction

With improving living standards and increasing life expectancy, many developing countries face the challenge of managing both an epidemiological (Gribble and Preston, 1993; Omran, 1971) and a demographic transition. As the epidemiological transition takes place, the relative contributions of communicable and non-communicable diseases to death rates, as well as to the burden of disease (loss of healthy life from death and disability) change. The major causes of death and of the burden of disease move from infectious diseases and undernutrition to non-communicable diseases such as cardiovascular disease, diabetes, and over-nutrition. An epidemiological transition combined with an aging population means that countries have the challenge of managing health care for a growing elderly population that suffers from traditional communicable diseases as well as chronic degenerative diseases. This paper investigates whether the Mexican conditional cash transfer (CCT) program, initially known as Progresa and now called Oportunidades, helped with this challenge by reducing elderly mortality.

Mexico is an example of a middle-income country that is fairly advanced in the epidemiological transition. In 1991, non-communicable diseases already accounted for 47% of the burden of disease, communicable diseases for 32%, and the rest was attributable to injuries and accidents (Lozano et al., 1995). By 2004, non-communicable diseases dominated, increasing to 68% of the burden of disease and 75% of total deaths, while communicable diseases had fallen to 14% of deaths and 18% of the burden of disease (Stevens et al., 2008). In addition, infectious diseases and malnutrition went from being 3 of the top 10 causes of death (respiratory infections, acute diarrhea, malnutrition) in 1991, to being only 1 of the top 10 causes of death (respiratory infections) in 2004. However, the speed of the transition differs by region in Mexico, with communicable diseases still accounting for up to a third of the

☆ We thank Brian Cadena, Dick Jessor, Terra McKinnish, Duncan Thomas, and anonymous reviewers for their helpful comments.
* Corresponding author at: Institute of Behavioral Science, University of Colorado at Boulder, 483 UCB, Boulder, CO 80309-0483, USA. Tel.: +1 303 492 0237; fax: +1 303 492 6924.
E-mail address: tania.barham@colorado.edu (T. Barham).

0304-3878/$ – see front matter © 2013 Elsevier B.V. All rights reserved.
http://dx.doi.org/10.1016/j.jdeveco.2013.08.002

burden of disease in 2004 in some of the poorer southern states (Stevens et al., 2008).

CCT programs are a popular social program in middle and low-income countries. They are usually thought of as a way to build the human capital of young children and break the intergenerational transmission of poverty. They pursue these goals through the provision of cash transfers conditional on beneficiaries engaging in positive behaviors such as children's attendance at school or regular health care visits (Fiszbein and Schady, 2009). However, the design of the Mexican CCT may also have led simultaneously to improvements in elderly health. In particular, the program required that all adults in a beneficiary household seek regular preventive health visits, and the health services provided addressed some of the health concerns of the elderly. A household member was also required to attend health education sessions to learn how to take better care of the health and nutrition needs of all family members, and not just the children (Adato et al., 2000). Moreover, the cash transfer increased the income of the entire household, allowing all household members to potentially improve their nutrition and buy more health inputs.

Progresa is one of the first large-scale CCT programs. It started in 1997 mainly in poor rural areas. By 2000, the program reached approximately 2.5 million families or about 40% of rural families (Coady, 2000). Subsequently the program expanded to include more urban areas, and served about 5 million households by 2007 (Adato and Hoddinott, 1999). Similar types of programs have been implemented in more than 30 middle- and low-income countries (Fiszbein and Schady, 2009). A prominent feature of Progresa is the randomized evaluation built into the design of the program. While many studies on Progresa take advantage of the data collected as part of the randomized evaluation, the sample size is insufficient to estimate the impact of Progresa on an important health indicator – elderly mortality – and does not contain information on cause of death. Due to these limitations, previous research has not examined the effect of Progresa on elderly mortality.

We exploit the phasing-in of Progresa throughout Mexico over time and location, and take advantage of the high quality death certificate data available in Mexico to estimate the effect of Progresa on elderly mortality for those aged 65 and older. Similar to Barham (2011), we use the percentage of households receiving Progresa transfers in a given year and municipality as the treatment variable and compare municipalities phased-in during 1998 to those in 1999 since they are the most similar to each other. Using a municipality-level dataset covering the period 1992–2002, and a municipality and time fixed-effects model, we show that Progresa led to a 4% reduction in the average, municipality-level elderly mortality rate; the effects are similar for men and women. Results by cause of death reveal that Progresa was successful at reducing deaths related to infectious diseases, diabetes, and nutrition and anemia, but not those related to transportation accidents, as would be expected given the program interventions. Given that the empirical model uses the second lag of the treatment variable, the results focus on the effect of Progresa through 2000, when the program mainly operated in poorer rural areas. We do not examine the effect of the program expansion into more urban areas after 2000 since the effect of Progresa on elderly health during this time period is confounded by the introduction of health reform. The health reform included an increase in public spending on health and a new public health insurance scheme, *Seguro Popular*, which benefited a similar population to that served by the CCT program (Frenk et al., 2006).

Previous research on the effects of CCTs on adult health and especially those 65 and older is limited, but the research that does exist focuses on Progresa and demonstrates that Progresa had short-run effects on possible pathways influencing health outcomes which suggest that it is reasonable to expect that the program led to a reduction in elderly mortality. For example, beneficiaries aged 50 or older experienced a 60% increase in health visits in the previous two months, a reduction in self-reported sick days, an increase in the number of days they were able to carry out normal activities, an increase in the proportion reporting that they could carry out vigorous activities, and a reduction in the proportion reporting high blood pressure (Behrman and Parker, 2011; Gertler and Boyce, 2001). Fernald et al. (2008) also show that the program resulted in a lower prevalence of uncontrolled hypertension among people aged 30–65 years old.[1] However, this is the first paper which demonstrates that Progresa has indeed reduced elderly mortality.

The rest of the paper proceeds as follows. Section 2 summarizes the Progresa program and the pathways through which program interventions may affect elderly mortality; Section 3 describes the data; Section 4 lays out the identification and estimation strategy; the findings are discussed in Section 5; and Section 6 concludes.

2. The Progresa program

2.1. Background

Progresa was introduced in 1997, and between 1997 and 2000 gave preference to rural localities since poverty was highest in these areas, though a limited number of urban localities were included. The program was expanded starting in 2001 to include more urban areas. The program was originally designed to alleviate short-term poverty and to break the intergenerational transmission of poverty by improving the health and development of children. A secondary objective of the program was to improve adult health (Fernald et al., 2008). Progresa combines two traditional methods of poverty alleviation: cash transfers and free provision of health and education services. Conditioning receipt of the cash transfers on children attending school, all family members obtaining regular preventative health care, and at least one family member attending health education training sessions, links the two objectives. Therefore, the income transfer not only relaxes the household budget constraint, but also helps improve the utilization of health and education services. There are separate transfers provided for the health and education conditionalities, so receiving the health transfer does not depend on the family also meeting the education conditionalities. The conditionalities were provided to a designated woman in the beneficiary household. Together, these conditionalities led to an increase in average beneficiary income levels of 22% in rural areas (Parker and Teruel, 2005).

2.2. The Progresa health and nutrition component

The health component of Progresa was designed to address health issues of all members of the family. The conditionalities required that all adults in the household, including senior citizens, have one preventative health check-up a year, and that at least one family member attend regular health education training sessions (Adato et al., 2000). Nutritional supplements were also given to mothers and young children. Transfers based on the health conditionality were paid every two months and were only paid if all the members of the beneficiary households attended the required health care visits and health education training sessions for that two-month period. Health clinics were required to provide a minimum package of services to ensure a basic quality of care. This package did not cover all types of health issues faced by families or the elderly but did include: family planning; education on basic sanitation, and accident prevention; prenatal, childbirth

[1] Fernald et al. (2008) found significant reductions in the body mass index and prevalence of obesity due to the program, but results are not significant once covariates are included.

and puerperal care; growth monitoring; vaccinations; anti-parasite treatment; and prevention and treatment of diarrhea, respiratory infections, tuberculosis, high blood pressure, and diabetes (Adato et al., 2000). The health package was expanded with the introduction of health reform in 2003 to cover, among other health issues, some cancers, cardiovascular problems, and dialysis (Knaul et al., 2005). However, this came after the time period covered in this study.

Since it was expected that health care utilization would rise as a result of the program, Progresa coordinated with other government ministries to increase the health care supply to ensure that the quality of health care in program areas did not deteriorate. The improvement in supply included the use of mobile clinics and foot doctors to reach marginalized communities that did not have access to permanent health clinics. Despite improved access to health care, many health services were still out of the reach for the poor. This is because the health care system available to the poor during this time period relied heavily on out-of-pocket spending and these costs were often beyond the reach of the poor; additionally, coverage of service was not comprehensive as described above in the basic package of services (Knaul et al., 2006).

2.3. Program targeting, phase-in, and take-up

For logistical and financial reasons, Progresa rolled out over time across localities (Skoufias et al., 1999). Progresa used a two-stage process to identify eligible households. In 1997, localities were selected based on a marginality index,[2] access to primary and secondary schools and to a permanent health care clinic.[3] The program used population density data and information on the proximity of localities to each other to determine the geographic isolation of the locality. Any locality with less than 50 inhabitants, or that was determined to be geographically isolated, was excluded from the program. In 1998, the condition that the locality has to have access to a permanent health care clinic was relaxed when mobile clinics and foot doctors became available. Some localities that were eligible but previously excluded due to geographic isolation were also incorporated into the program.[4]

Once localities were selected, beneficiary households in each community were identified based on their poverty status as determined by household income and characteristics collected in a census of the program localities. Only those classified as poor were eligible for benefits. Recertification of eligibility was supposed to take place after three years of participation (Coady, 2000). As a result of this process, the program covers a different percentage of the population in each locality. Households were informed that they were eligible using door-to-door methods. Research in areas where the randomized intervention took place report that take-up of the program was remarkably high at 97% (Gertler and Boyce, 2001).

Fig. 1 shows the expansion of the program over time across localities.[5] The program started in a limited number of localities in 1997, with the first transfers being provided between September and October of 1997. In 1998, the program was greatly expanded, reaching around 35,000 localities in all but two states, and over 95% of these localities were rural localities. Not all localities were incorporated in the same month in 1998; the expansion was spread out during the year. For this reason, some beneficiaries received transfers starting in February 1998, whereas others had to wait until September or October of 1998. There was another expansion of localities and beneficiaries in 1999. The next major expansion took place in 2001 and 2002 with the incorporation of more urban localities.

2.4. Program mechanisms to reduce elderly mortality

The Progresa program could affect elderly mortality rates through a number of mechanisms. The health conditionalities (annual preventive visit and health education sessions) could have improved health since many of the services in the basic health services package addressed common health problems of the elderly in Mexico. In particular, the package included services for two risk factors for cardiovascular disease, hypertension and diabetes (WHO, 2011), respiratory infections, and recipients received more regular monitoring of their health. In addition, the health education trainings covered health issues relevant to the elderly, including nutrition (eating well for good health), accident prevention, and chronic disease detection and prevention (Adato et al., 2000). While not specifically documented, the health education training sessions or doctors at the preventive check-ups may have provided specific information on important health behaviors for reducing chronic diseases like cardiovascular diseases, such as the importance of exercise, diet, and controlling tobacco and alcohol use (WHO, 2011).

The cash transfer may also have improved elderly health by enabling beneficiary households to buy more nutritious foods and spend more money on health inputs. However, it is possible that the income transfer may have exacerbated problems of overweight and obesity and increased the risk of death if households increased consumption of non-nutritious foods or calories. This possible negative effect may have been tempered by the health education training sessions, since healthy eating behaviors were promoted during these sessions (Adato et al., 2000). Indeed, prior research shows that Progresa led to a 6.4% increase in household caloric intake, but that the increase was greatest for higher quality foods such as vegetables, fruits, milk, eggs and meat (Hoddinott and Skoufias, 2004).

Finally, the schooling conditionality could also affect the health of elderly household members if the required attendance of children at school provided the elderly with more time to engage in positive health behaviors or required them to do more exercise due to the need to perform household chores usually undertaken by the children. Overall, these are plausible pathways for the Progresa CCT to have impacted the mortality of Mexican elders.

3. The data

We construct a municipality-level data set covering the 1992–2002 time period using vital statistics information, Progresa administrative records, and census data.[6] We examine the effect of the program on the elderly mortality rate (EMR), i.e., the number of deaths per 1000 population, for three age groups for both sexes: those aged 65 and older (EMR 65+), those aged 65–74 (EMR 65–74), and those aged 75 and older (EMR 75+).

The EMR is constructed at the municipality level by sex using vital statistics data on deaths and census data on population. The vital statistics data are from a nation-wide database containing information on every certified death in Mexico and were provided by the Mexican Ministry of Public Health. These data have municipality level identifiers and are not publicly available. The population data are from the 1990, 1995, 2000, and 2005 censuses and are publicly available from the

[2] The marginality index included the following variables: the literacy rate; percent of dwellings with running water, drainage, and electricity; average occupants per room; percent of dwellings with a dirt floor; and percent of labor force working in the agriculture sector.

[3] A locality was considered to have access to a health care clinic if the clinic was either in the locality or in a neighboring locality at most 15 km away (Skoufias et al., 1999).

[4] See Skoufia et al. (1999) and Coady (2000) for more details on program targeting.

[5] These numbers are based on Progresa administrative data provided to the first author by the central Oportunidades administrative office in 2004 and represent localities that had at least one household registered for the program by December of each year.

[6] Localities in Mexico are grouped into municipalities. The 2000 census recorded that there were 199,391 localities in 2445 municipalities in Mexico.

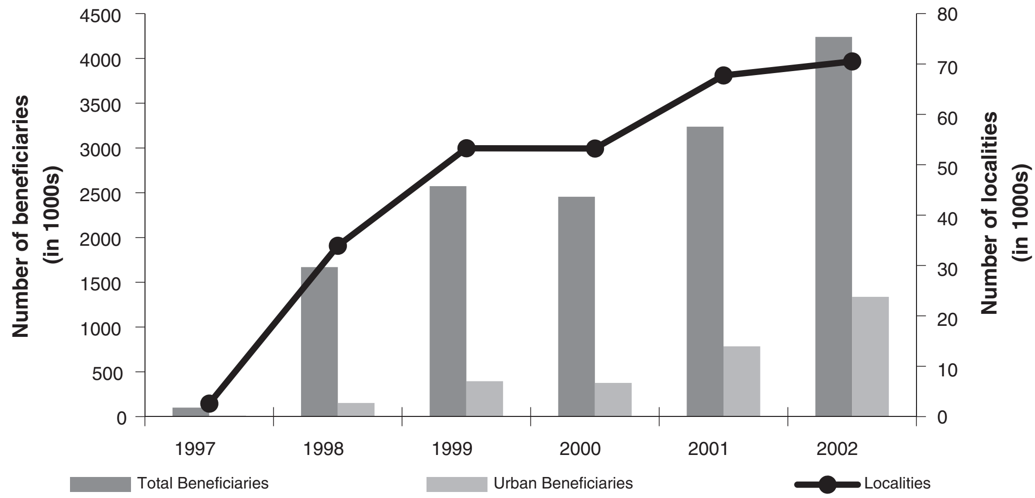

Fig. 1. Trends in Progresa beneficiary families and localities.

Mexican Statistical Agency (INEGI) at the municipality level by age and sex. Linear interpolation was used to determine population values for the years that the data are not available.[7]

Mexico has a long-standing vital registration system that has met the World Health Organization's (WHO) international standards since the 1950s and is ranked in the top 20 in the world for its quality and completeness (Braine, 2006; Mathers et al., 2005). Despite this high rating, the quality of the cause of death data does vary depending on whether the death is certified by medical personnel versus a lay-person authorized by the Ministry of Health, and under-reporting remains an issue (Braine, 2006; Hernández et al., 2011). The percentage of elderly deaths in rural areas certified by a medical practitioner is high but did rise from 87% in 1992 to 92% in 2002. To account for this improvement in cause of death reporting during the study period, we control for the percentage of deaths certified by a medical practitioner as a robustness check when examining the effects of the program by cause of death. If a cause of death is not determined, it is recorded as ill-defined. The percentage of elderly deaths that are ill-defined remained fairly stable over the study period at 6%. To determine if Progresa has an impact on ill-defined causes of death, we treat the ill-defined deaths in a similar manner to other causes of death. Finally, we discuss how changes in under-reporting may bias the results in Section 5.3.

The intensity of treatment indicator, referred to as *program intensity*, is the ratio of the number of households receiving Progresa benefits to the total number of households in a municipality. This variable ranges from zero to one, where one indicates that Progresa covers all households in a municipality.[8] The numerator of *program intensity* is created using Progresa administrative data on the number of households registered in a locality for the program in December of each year and is available since the program started in 1997. The data are aggregated to the municipality level using municipality identifiers in the administrative data. These data are not publically available, but provided by the central Progresa administrative office. The denominator of program intensity is created using INEGI census data on the number of households in a municipality for 1990, 1995, 2000, and 2005. The number of households for each year between 1992 and 2002 is linearly interpolated.[9] While the study covers the period 1992–2002, variation in program intensity is only used until 2000 since the estimation equation uses the second lag of program intensity.

The municipality characteristics used as controls in the robustness analyses are created from locality level census data from INEGI for the years 1990,[10] 1995, and 2000. These variables are aggregated to the municipality level for those localities that received Progresa benefits before 2001 to better approximate change in characteristics of Progresa areas of municipalities over time. Again, linear interpolation is used to obtain values for the years that the data are not available.[11] The variables created using these data are: municipality density (population per square kilometer); the percentage of households with no piped water, with no electricity, with no wastewater disposal,[12] and with a dirt floor; the percentage of the population over 14 that are illiterate; the percentage of the population over 4 who speak an indigenous language; the percentage of employed working in the primary sector; and, the number of occupants per household.[13]

Yearly health supply data are from the Ministry of Health and Instituto Mexicano del Seguro Social (IMSS)-Oportunidades for the years 1992–2002 and are not publically available. These data do not include information on the total supply of health care in Mexico, as they only cover the providers that care for Progresa beneficiaries.

[7] Linear interpolation of the population data will not be an accurate estimation if there are sizable migration flows of the elderly that are correlated with Progresa. Given the short duration of the analysis, and that migration of the elderly is low, any measurement error should be minimized. In addition, Stecklov et al. (2005) find that for people under age 60, Progresa had no effect on domestic migration.

[8] There are a handful of observations that have values of program intensity slightly greater than one. These are mostly municipalities in which all localities in the municipality participated in the program. The ratio may be greater than one due to measurement error arising from the linear interpolation of the population data or due to locality boundaries being unclear when the administrative data was collected. We top code these values at one, but the results are same if we set them to missing, or leave them unchanged.

[9] If Progresa induced elderly individuals who were previously living on their own to move into a household with other family members, then the changes in number of households may not be accurately captured by linear interpolation. Given that the proportion of elderly living on their own prior to the program in 1994 was 7% in Mexico (Palloni, 2002), and that Progresa had little effect on the living arrangements of those aged 70 and older (Rubalcava and Teruel, 2006), any measurement error will be small.

[10] The 1990 locality data was matched by name to the 1995 and 2000 data. Locality names and codes changed over time and these changes were incorporated when identified.

[11] Due to the linear interpolation, if there is a large non-linear change in the covariates that is correlated with the treatment variable, the results could be biased. Given that Progresa localities were determined prior to the program based on pre-program characteristics, this possibility is minimized.

[12] No wastewater disposal means that the house does not have a drainage system that removes wastewater via the public sewage network, a septic take, creek, river, or some other water body.

[13] The percentage of households with a dirt floor and percentage employed working in the primary sector are only available in the 1990 and 2000 censuses.

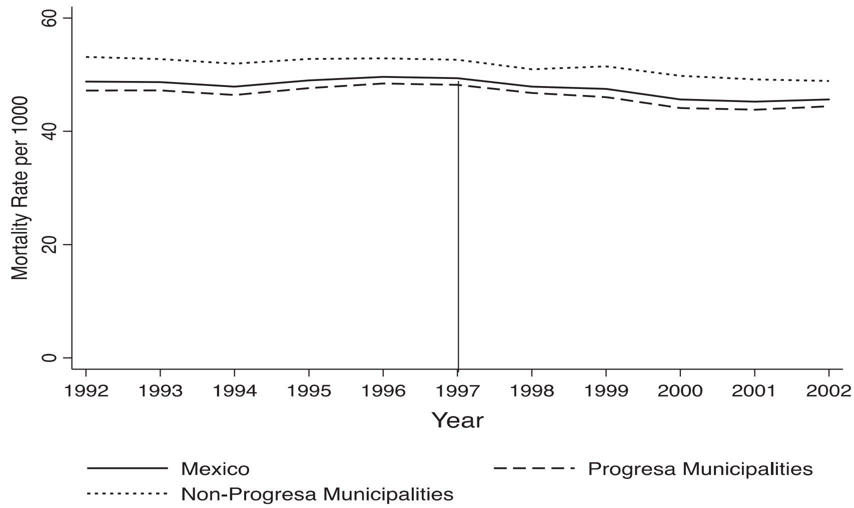

Fig. 2. Trends in EMR 65+ by Progresa Status.
Note: Progresa municipalities include municipalities incorporated into Progresa between 1997 and 2000, and differ from the analysis sample which only includes municipalities incorporated in 1998 and 1999. Non-Progresa municipalities are all other municipalities. EMR 65+ is the mortality rate for people aged 65 or older per 1000.

These data are at the locality level but aggregated to the municipality level and then converted to measures per 1000 population using census population data. The data are used to control for the number of clinics, hospitals, mobile clinics, health brigades, medical residents, as well as doctors and nurses in contact with patients in a municipality. To help control for the phase-in rule we also use these data to determine the proportion of Progresa localities in a municipality that have a permanent health facility (clinic or hospital) in a given year.

Using these data sources, we construct a municipality-level panel dataset for the years 1992–2002.[14] Municipality boundaries were redefined during this time period. In order to make a consistent panel of municipalities from 1992 to 2002, municipalities that were split in a particular year are amalgamated. This results in a balanced panel of 2376 municipalities each year. The research design only uses data on municipalities that were incorporated into Progresa in 1998 and 1999, leaving a panel of 1961 municipalities.

3.1. Summary statistics

Fig. 2 shows the trends in the elderly mortality rate for the 65 plus age group (EMR 65+) from 1992 to 2002 for all of Mexico, for municipalities that were incorporated into Progresa prior to 2000 (Progresa municipalities), and for those that were incorporated later or not at all (non-Progresa municipalities).[15] The general trend in EMR is similar for Mexico as a whole and in Progresa municipalities. In Progresa municipalities, the EMR was 49 deaths per 1000 population prior to Progresa in 1996, and was fairly stable in the five years prior to the program. The EMR declined between 1996 and 2000 to 44 deaths per 1000 in 2000. A similarly sized decline was not observed in the non-Progresa municipalities between 1996 and 2000. These trends show the raw data and do not mimic the research design, but demonstrate that there was a decline in the EMR over the period of the program in municipalities where Progresa operated.

Table 1 presents mean municipality EMR for municipalities that are used in the analysis sample — municipalities incorporated into Progresa in 1998 and 1999 only. We present the data for 1997 (prior to sample municipalities being incorporated into Progresa) and 2002 (the last year of data). Panel A highlights that EMR 65+ declined on average by 5 deaths per 1000 population aged 65 and older between 1997 and 2000. Panel B disaggregates the EMR by selected causes of death, where the first five causes are the ones Progresa was more likely to affect. While the EMR for those aged 65 and older declined over time for most causes of deaths that Progresa was likely to affect, diabetes related deaths in fact increased on average between 1997 and 2002. There was little change over time in deaths due to causes that Progresa was less likely to affect, such as cancer or transportation accidents.

4. Identification strategy

We estimate the impact of Progresa on elderly mortality at the municipality level, rather than the locality level – the level at which it was rolled out – due to data limitations. As a result of the unit of analysis being the municipality level, the treatment variable, *program intensity*, combines three sources of variation. The first is the variation in treatment status across municipalities over time. The second is the variation in the number of localities covered by the program over time within a municipality. Finally, the third source results from the percent of households benefiting from the program differing between localities.

4.1. Sources of variation

Municipalities were incorporated into the program over time starting in 1997. A municipality is defined as being incorporated into Progresa the year the first locality was phased-in in that municipality. Following Barham (2011), we compare municipalities phased-in during 1998 to those phased-in during 1999 since mortality rates between these sets of municipalities are more similar to each other than to municipalities phased-in either earlier, in 1997, or later. This is not surprising since, as described in Section 2.3, the requirement that the locality had to have access to a permanent health care clinic was relaxed in 1998 (so localities phased-in before 1998 had better health care access) and municipalities

[14] Mortality data is available starting in 1990 and health supply data in 1992. We take advantage of these two extra years of mortality data in the robustness check which uses lagged mortality. Results are robust to which pre-intervention years (1990–1996) are included in the analysis.

[15] The EMR is lower in Progresa municipalities than for Mexico as a whole. However, not all localities within the municipality received Progresa, and those that did were worse off (see Table 1 in the on-line appendix). As a result, the EMR may be higher in Progresa areas of a municipality than for the municipality as a whole.

Table 1
Means of EMR 65+ — analysis sample.

	1997		2002	
	Mean	SD	Mean	SD
Panel A: all causes of mortality				
Both sexes	47.5	(15.34)	42.5	(13.52)
Males	49.3	(21.95)	44.4	(17.63)
Females	46.0	(19.66)	40.7	(17.11)
Panel B: for both sexes for selected causes of death				
Infectious diseases	1.6	(2.92)	1.2	(2.24)
Heart, stroke and hypertension	14.4	(8.51)	13.0	(7.81)
Diabetes	3.0	(3.05)	4.1	(3.57)
Respiratory	5.9	(4.76)	4.8	(4.14)
Nutrition and anemia	3.6	(5.64)	2.9	(4.80)
Cancer	5.1	(4.19)	5.1	(3.87)
Transportation accidents	0.2	(0.73)	0.3	(0.75)
Ill-defined	2.9	(6.92)	2.9	(5.78)
All other causes	10.9	(6.86)	8.2	(5.23)

Notes: N = 1961. EMR 65+ is the mortality rate for people aged 65 older per 1000. The analysis sample includes municipalities incorporated into Progresa in 1998 and 1999.

incorporated after 1999 tended to be more geographically isolated or urban.

To identify the program effect using this municipality variation, we use municipalities yet to be treated as comparison municipalities. The identifying assumption in this case is that the trends in mortality between municipalities incorporated in 1998 and 1999 would be the same in the absence of the program. Although it is not possible to test this assumption directly, we test that the differences in pre-intervention means by year between municipalities that were incorporated in 1998 and 1999 are similar, after controlling for municipality fixed-effects and present the differences in Table 2. The differences in means are fairly small, and not statistically significant between the two groups for the years 1992 to 1996 providing some confidence that the municipalities incorporated in 1998 are similar to those incorporated in 1999. In addition, as a robustness check we lag the dependent variable 3 years to verify that the treatment variable is not correlated to mortality trends prior to the program.

Another source of variation used to identify the program impact is the phasing-in of Progresa localities, within a municipality, over time. This source of variation arises because localities within a municipality were incorporated into the program in different years. Results may be biased if the trends in mortality in localities that were phased-in during different years within a municipality are not similar. Since mortality data is not available at the locality level, following Barham (2011), we use locality level census data to show that the pre-intervention trends in locality characteristics are fairly similar between localities phased-in in different years. Table 2 of the on-line appendix presents evidence similar to Barham (2011) on pre-intervention trends in locality characteristics for the analysis municipalities — municipalities incorporated in 1998 and 1999. The locality characteristics include literacy; access to services such as electricity, piped water, and wastewater disposal; household size; and the ethnicity of the population. These variables are proxies for income and education, two important determinants of elderly mortality. The first row of the table displays the change in means between 1990 and 1995 for localities that were incorporated into the program in 1998. The subsequent rows show how these changes differ between localities that were incorporated into the program in 1998 versus later years (1999, 2000, and 2001), and indicate whether the changes are statistically different between the two groups. The differences in means are adjusted for municipality fixed effects, mimicking the regression analysis.[16] The changes in means are statistically different for many variables. This is not surprising given the large sample size (approximately 50,000). The magnitudes are, however, arguably small.

We examine two other measures of the size of the difference since statistical tests, such as t-tests, are affected by sample size. First, we examine normalized differences, that is, the difference in the changes between groups divided by the standard deviation for the sample, since these are not affected by sample size. As a rule of thumb, normalized differences greater than 0.25 may lead to sensitive results (Imbens and Wooldridge, 2009). The normalized differences are less than a quarter and all but three are less than 0.1. Second, the difference in the change in means is compared to the difference in quartile cut-off values (i.e., fourth quartile cut-off minus third quartile cut-off, third quartile cut-off minus second quartile cut-off, etc.) for each variable for 1990. The differences in the change in means are smaller than the non-zero differences between the quartiles of the distribution, indicating that they are not large enough to move a locality to a different part of the distribution. Lastly, these variables are included as controls in the regressions as a robustness check that any differences in time varying observables are not biasing the results.

Finally, there is variation in the treatment variable, *program intensity*, across localities. This variation is in part due to poverty targeting by the program within the locality, and may mean that the percent of households covered by the program in a given locality is endogenous. The variation across localities could also be a result of differential program take-up between localities. The program take-up rate is high in areas that were part of the randomized evaluation, 97%, so selection into the program is likely to be the smaller issue. To address these potential issues, we create a treatment variable, *locality intensity*, that measures the percentage of localities that have Progresa beneficiaries in a municipality, and consequently does not depend on the number of beneficiary households. It is possible that locality intensity is correlated with poverty at the municipality level. However the inclusion of municipality fixed-effects will control for non-time varying differences in poverty between the municipalities.

Using these sources of variation, the treatment variable, *program intensity*, ranges from zero to one, where one indicates that 100% of households in a municipality were covered by Progresa. Given that the regression model uses the second lag of program intensity, the analysis only uses variation in this variable until 2000. The average program intensity for municipalities phased-in during 1998 started at 28% in 1998 and reached 35% by 2000. In municipalities incorporated in 1999, the average program intensity was around 27% in both 1999 and 2000.

Table 2
Difference in mean EMR 65+ between municipalities incorporated into Progresa in 1998 and 1999, by year.

Year	Difference in means (incorporated 1999 − incorporated 1998)
1992	−1.10
	(0.89)
1993	0.33
	(0.83)
1994	−0.35
	(0.87)
1995	−0.85
	(0.83)
1996	0.31
	(0.78)

Notes: These results are the differences in means controlling for municipality fixed effects. Standard errors are in parentheses and are clustered at the municipality level. EMR 65+ is the elderly mortality rate for those aged 65 and older per 1000.

[16] The magnitude of the differences in *locality characteristics* between localities phased-in in 1999 and 1998 in Table 2 of the on-line appendix, is similar to the magnitude of the differences in *municipality characteristics* between municipalities incorporated in 1999 and 1998 (results not reported).

Table 3
Impact of Progresa on EMR 65+.

	Both sexes	Male	Female	Robustness checks — both sexes				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
2nd lag of program intensity	−6.37	−6.42	−6.46	−0.98		−6.13	−5.14	−6.59
	(1.04)	(1.42)	(1.31)	(1.07)		(1.07)	(1.51)	(1.06)
2nd lag of locality intensity					−4.91			
					(0.88)			
N	21,571	21,571	21,571	19,610	21,571	21,571	21,571	21,571
R^2	0.36	0.24	0.27	0.37	0.35	0.36	0.44	0.36
Mean DV	47	48	46	47	47	47	47	47
EMR 65+ lagged 3 years	N	N	N	Y	N	N	N	N
Municipality characteristics	N	N	N	N	N	Y	Y	N
Municipality time trend	N	N	N	N	N	N	Y	N
Health supply	N	N	N	N	N	N	N	Y

Notes: All regressions include municipality and time fixed effects. Standard errors are in parentheses and are clustered at the municipality level. Mean DV is the mean of the dependent variable in the pre-intervention years. The municipality characteristics include controls for: population density; percentage of Progresa localities with access to a permanent health facility; the percentage of households with no piped water, with no electricity, with no wastewater disposal, and with a dirt floor; the percentage of the population over 14 that is illiterate; the percentage of the population over 4 that speaks an indigenous language; the percentage of employed individuals working in the primary sector; and the number of occupants per household. Health care supply controls are per 1000 population and include: health clinics, hospitals, mobile health clinics, health brigades, doctors, residents, and nurses.

4.2. Empirical model

The municipality-level average treatment effect is estimated using the following equation:

$$(2)\ EMR_{mt} = \alpha_t + \tau_m + \beta program\ \ intensity_{m,t-2} + \varepsilon_{mt},$$

where *EMR* is the elderly mortality rate in municipality *m* in time *t* and *program intensity* is the measure of the intensity of treatment (percent of households receiving Progresa transfers in a municipality).[17] The second lag of program intensity is included since some new beneficiaries started receiving benefits in the second half of the year, and it can take some time for health care visits to have an effect on health.[18] Year fixed effects, α_t, are included to control for general time trends common to all municipalities, and municipality fixed effects, τ_m, are included to capture time-invariant municipality-level unobservables. The error term is clustered at the municipality level to account for likely intracluster and serial correlation, as well as the heteroskedasticity that is inherent in models with aggregated data.

The effect of Progresa is given by β and shows the effect of the program if the program intensity rose from zero to one (i.e., 100% of households in a municipality are Progresa beneficiaries). To estimate the average municipality-level program effect we multiply β by the average of the program intensity in 2000, 0.32. The estimate of the treatment effect will be unbiased if there are no unobserved time-varying municipality characteristics that are correlated with the treatment variable. The similarity of the trends in the mortality and locality characteristics for the quasi-experimental treatment and comparison groups prior to the program provide some confidence that the unobservables are not biasing the results. However, to further test that time varying unobservables are not biasing the results we include municipality controls and individual municipality time trends in the regression model as a robustness check.

5. Results

5.1. Program impacts

Table 3 presents the effects of Progresa on the EMR for those aged 65 and older. Column (1) presents the effect of the program after controlling for year and municipality fixed effects only.[19] The point estimate on lagged program intensity is statistically significant at the 1% level and is 6.37. This means that a one percentage point increase in program intensity leads to a 0.0637 per 1000 individual decrease in the elderly mortality rate for people aged 65 and older. At the municipality level, the proportion of households covered by the program reached an average of 32% in 2000, so the average municipality program effect represents a decline of approximately 4% in the EMR with respect to the pre-program level of 47. Alternatively, a standard deviation increase in program intensity led to a reduction in the EMR of 1.47 deaths per 1000, which is a 3% decline over the pre-program EMR. Columns (2) and (3) show that the program effect for males and females are similar at 6.42 and 6.46 respectively.

Results disaggregated by age, 65–74 and age 75 plus, are in the on-line appendix Table 3. The average municipality program effect is higher for the 65–74 year olds than for the 75 plus group, at 5% compared to 3%.

5.2. Robustness checks

This section explores a number of threats to the validity of the estimates (Table 3 columns 4–8). Given the similarity in results between males and females, we do not present results by gender in this section.

First, as discussed in Section 4.1, to test that pre-existing trends are not related to the treatment variable variation, the dependent variable is lagged three years. The coefficient on the treatment variable in column (4) is −0.98.[20] It is small and not significantly different from zero, suggesting that pre-existing trends are not biasing the results.

Second, Section 4.1 also highlighted the possible endogeneity in the determination of the percentage of households covered by the program in a given locality. To address this potential bias a new treatment variable is created which is not based on the number of households in a municipality, *locality intensity*, which is defined as the percentage of localities that have Progresa beneficiaries in a municipality. The point estimate on the treatment variable in column (5), −4.91, shows a slightly smaller

[17] This specification assumes that the treatment effect is linear. Examining the treatment effect by 0.2 bins of program intensity shows this assumption holds.

[18] Specification tests on the number of lags of program intensity support using the second lag.

[19] To the extent that non-eligibles (non-poor in a locality) benefit from the improved supply of health care services or the health education program, the program effects may be over-estimated. Bobonis and Finan (2002) find no evidence of health spillover effects on the non-eligibles in Progresa localities using the Progresa randomized evaluation database.

[20] The expansion of the supply of health care services prior to the program could have affected pre-program trends.

Table 4
Impact of Progresa on EMR 65+ by cause of death.

	Both sexes	Male	Female	Both sexes	Male	Female
	Infectious disease			Cancer		
2nd lag of program intensity	−1.70	−1.27	−2.11	−0.04	−0.46	0.32
	(0.23)	(0.31)	(0.28)	(0.25)	(0.35)	(0.34)
R^2	0.25	0.18	0.19	0.26	0.22	0.15
Mean DV	2.5	2.6	2.4	4.8	5.1	4.5
	Heart disease, stroke, and hypertension			Transportation accidents		
2nd lag of program intensity	−0.31	−0.76	0.12	−0.01	−0.02	−0.00
	(0.58)	(0.74)	(0.76)	(0.06)	(0.10)	(0.06)
R^2	0.31	0.22	0.23	0.14	0.12	0.10
Mean DV	13.9	13.6	14.4	0.2	0.3	0.1
	Diabetes			Ill-defined cause of death		
2nd lag of program intensity	−1.03	−0.98	−1.10	−1.34	−0.86	−1.80
	(0.18)	(0.26)	(0.28)	(0.48)	(0.52)	(0.61)
R^2	0.39	0.27	0.29	0.41	0.31	0.34
Mean DV	2.6	2.1	3.1	3.1	2.8	3.5
	Respiratory			All other causes		
2nd lag of program intensity	−0.28	−0.23	−0.34	−0.01	−0.52	0.49
	(0.34)	(0.45)	(0.43)	(0.41)	(0.66)	(0.47)
R^2	0.25	0.20	0.19	0.22	0.19	0.17
Mean DV	5.5	5.9	5.1	10.7	12.5	9.1
	Nutrition and anemia					
2nd lag of program intensity	−1.66	−1.32	−2.05			
	(0.46)	(0.53)	(0.55)			
R^2	0.29	0.20	0.24			
Mean DV	3.5	3.2	3.8			

Notes: N = 21,571. Following column 1 in Table 3, all regressions include municipality and time fixed effects. Standard errors are in parentheses and are clustered at the municipality level. Mean DV is the mean of the dependent variable in the pre-intervention years.

reduction in deaths due to the program (3%), but again is not statistically different from column 1.

Third, we test whether the point estimates are sensitive to the inclusion of time varying municipality characteristics in column (6) by controlling for the phase-in rule (population density and the percentage of Progresa localities in a municipality with access to a permanent health facility in a given year), and variables that are likely correlated with poverty and mortality from the census. It is possible that these controls are endogenous if Progresa led to changes in these variables, so we include the controls only as a robustness check. To better approximate changes that took place in Progresa areas we include municipality controls that only reflect characteristics for localities that received Progresa benefits before 2001, rather than including municipality controls that reflects characteristics of all the localities in the municipality. The point estimate on lagged program intensity is −6.13. The similarity in results with and without observable time varying characteristics provides some confidence that differences in the time varying observables are not biasing the results.[21]

Fourth, to further test that unobservable time varying characteristics are not biasing the results, individual municipality time trends are included in column (7). The program effect is slightly smaller at −5.14 but not statistically different from column (1), and shows a program effect of 3.5%.

Fifth, localities phased-in during 2000 were generally likely to be more geographically isolated and worse off than those phased-in earlier. While the research strategy does not include municipalities incorporated in 2000, there are some localities that were phased-in during 2000 in the municipalities in the sample. The proportion of localities phased-in during 2000 is small (less than 1% of localities in sample municipalities). However, in order to exclude these localities we drop the 2002 data. This excludes the use of variation in program intensity from 2000 because we use the second lag of program intensity in the regression analysis. Again, the point estimate on the treatment variable excluding 2002 is similar to column (1) at −6.95 (result not reported).

Finally, it is possible that the program effects are partly a result of the expansion of health care services in rural communities that preceded Progresa. The supply of health care services was increased to ensure that the quality of services provided did not deteriorate with the increase in health care service utilization resulting from the program (Bautista-Arredondo et al., 2006), and also that beneficiaries could meet the health conditionalities. Health care was expanded in both treatment and comparison municipalities (i.e., municipalities incorporated in 1998 and 1999), but if the timing of expansion varied it is then difficult to determine whether the reductions in mortality are due solely to the CCT program. Health care supply variables are likely to be endogenous but are included in column (8) to examine whether the increase in the supply of health care services is a mechanism through which Progresa reduced mortality. The point estimate on the second lag of program intensity for the model including health care supply controls, at −6.59, is not significantly different from the point estimate for the model without these controls in column (1), providing some evidence that changes in the supply of health care is not the mechanism driving the results.[22]

5.3. *Biases from under-reporting of vital statistics data*

Vital statistic data on deaths usually suffers from under-reporting. To the extent that under-reporting does not change over time within a municipality, it is controlled for by the municipality fixed-effects. Under-reporting of deaths in Mexico occurs if the time or money costs associated with reporting the death to a health official are too high (Braine, 2006). Progresa may actually lead to an increase in reporting of deaths among beneficiary families because all members of the family must

[21] If we use municipality controls that represent the whole municipality rather than just Progresa localities within the municipality the results are again similar (−6.39).

[22] Results are similar whether health care supply variables are lagged for one or two years.

Table 5
Impact of Progresa on EMR 65+ by pre-program mortality levels.

	Both sexes	Male	Female
2nd lag of program intensity	−3.44*	−3.69*	−3.17+
	(1.37)	(1.88)	(1.86)
2nd lag of program intensity ∗ high pre-program EMR	−7.42**	−6.91*	−8.25**
	(2.01)	(2.80)	(2.59)
R^2	0.38	0.26	0.29
Mean DV low pre-program EMR	41	42	39
Mean DV high pre-program EMR	53	54	52

Notes: N = 21,571. Following column 1 in Table 3, all regressions include municipality and time fixed effects. Standard errors are in parentheses and are clustered at the municipality level. High pre-program EMR are municipalities for which EMR 65+ was greater than the median EMR 65+ for the sample in 1995. Mean DV is the mean of the dependent variable in the pre-intervention years.

attend a regular preventive care visit or the family will not receive the cash transfer for the health component. So, it is in the financial interest of the family to report a death to the health official. As a result, it is possible that in areas where there are more Progresa beneficiaries there will be more deaths reported than before the program began, likely causing an under-estimate of the program effect.

5.4. Impact of the program on selected causes of deaths

To better understand the types of illnesses that Progresa was more effective at addressing, Table 4 presents the program impact on the EMR for those aged 65 and older by cause of death. We include 5 major cause of death categories that Progresa may have affected, based on services provided in the minimum package of health services and other program interventions (infectious diseases; heart disease, stroke and hypertension — core components of cardiovascular disease; diabetes; respiratory infections; and nutrition and anemia). We also examine the effect of the program on two causes of death that are less likely to be affected by Progresa: cancer, which is health related, and, deaths due to transportation accidents, which are non-health related. Finally, we examine the effect of the program when the cause of death was not determined (ill-defined cause of death), and include a category "all other causes", which includes deaths due to any cause that is not specifically reported in the table.

Results are presented for both sexes together as well as separately. We discuss the results for both sexes together unless the results differ by sex. There is a statistically significant reduction (at the 1% level) of deaths per 1000 of 1.70 due to infectious diseases (or 22%), 1.03 due to diabetes (or 12%), and 1.66 due to nutrition and anemia (or 15%). There is little effect of the program on respiratory infections or heart disease, stroke and hypertension. It is somewhat surprising that there is not a bigger effect on heart disease, stroke and hypertension given that the basic health package focused on two important risk factors, hypertension and diabetes. However, services to address hypertension as well as to promote the prevention of cardiovascular diseases, such as life style changes and statin therapy, were inadequate during this time period (Gómez-Pérez et al., 2010; Mexican Ministry of Health, 2006).

As expected, the point estimates for changes in death rates due to cancer and transportation accidents are small and insignificant at −0.04 and −0.01. To the extent that the unobservables driving the trends for these two causes of death are similar to those for the other causes of death, this provides some further evidence that trends in unobservables are not biasing the results.

The program did reduce the number of deaths that were ill-defined. It is unclear whether the decline in ill-defined deaths reflects a real reduction in deaths from ill-defined causes, or whether there were fewer deaths recorded as ill-defined because the quality of cause of death data improved with the program. However, both interpretations of the reduction in ill-defined deaths could potentially lead to an under-estimation of the program effect for the other causes of death. For example, if deaths due to diabetes are better diagnosed as a result of the program, then deaths that would have been recorded as ill-defined prior to the program, but were actually due to diabetes, may be recorded as due to diabetes after the program was introduced.

As discussed in Section 3, the quality of the cause of death information varies depending on whether the cause of death was certified by a medical practitioner or a lay-person authorized by the Ministry of Health. We control for the proportion of deaths in the municipality certified by a medical practitioner each year since this figure rose over time; the results are similar (results not reported).

5.5. Heterogeneity by pre-intervention mortality levels

It may be easier to reduce the mortality rate when it starts at a higher level, so we examine whether the impact of the program varies as a function of the pre-program elderly mortality rate. We interact the treatment variable and the time fixed-effects with an indicator variable that is set equal to one if the municipality had an elderly mortality rate in 1995 that was above the sample median for 1995. The findings in Table 5 indicate that the program led to a −3.44 decline in the EMR for those aged 65 and older in program areas that had a lower mortality prior to the program. The decline for municipalities that had higher initial EMR was larger by 7.42 deaths, and the difference is statistically significant.[23] Given a mean death rate of 53 per 1000 in the years prior to the program and an average program intensity of 32% in 2000 in municipalities that had higher initial mortality rates, the average program effect is slightly higher than for the sample as a whole, at 6%. Results by gender are again similar.

5.6. Heterogeneity of the treatment effect by pre-intervention characteristics

To provide some insight into whether or not certain characteristics of municipalities may have resulted in higher success rates, we examine whether there were heterogeneous program effect by the size of the municipality and various pre-intervention determinants of poverty and mortality. To create the pre-intervention characteristics for all variables except municipality size, locality data from the 1995 census are aggregated to the municipality level for all localities incorporated into Progresa by 2000.[24] Terciles of each characteristic are created so that higher terciles reflect a worse off state, and the third tercile is interacted with the treatment variable to compare those in the worse off tercile to the two better terciles. We use the total number of households in a municipality in 1995 to measure municipality size. Similarly to the other pre-intervention characteristics, we interact the third tercile with the treatment variable, where the third tercile indicates that a municipality has fewer households than the first and second terciles. We examine the heterogeneous effects for each variable in separate regressions reported in columns 1 to 8 in Table 6. To take into account the correlations among the variables, we include all variables in one regression in column 9.

The results indicate that the program effect is larger in smaller municipalities, and in areas that had less access to electricity and some kind of a wastewater disposal system prior to the program. Larger program effects in areas with less access to electricity and wastewater disposal systems may indicate that the program was simply more

[23] The relative ranking of the death rate at the municipality level does not necessarily correlate with the relative ranking of the death rates in localities in which Progresa operated in those municipalities.

[24] Data on the percentage of households with a dirt floor and the percentage of the rural population working in the primary sector is from 1990, since 1995 data are not available.

Table 6
Heterogeneity of the impact of Progresa on EMR 65+ by pre-intervention characteristics, both sexes.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
2nd lag of program intensity	−2.16	−5.98	−4.99	−3.23	−5.30	−5.35	−7.10	−5.64	−0.11
	(0.98)	(1.24)	(1.31)	(1.27)	(1.46)	(1.30)	(1.43)	(1.28)	(1.81)
Interaction of 2nd *lag of program intensity with third tercile indicator variable of*:									
Number of households	−7.04								−6.86
	(1.51)								(1.57)
Percentage of households in Progresa localities:									
No piped water		−0.36							0.18
		(1.49)							(1.63)
No electricity			−2.76						−4.21
			(1.55)						(1.65)
No wastewater disposal				−5.14					−4.01
				(1.50)					(1.60)
Dirt floor					−1.20				0.73
					(1.54)				(1.70)
Percentage in Progresa localities:									
Speak indigenous language						−1.17			−0.40
(% population > 4)						(1.49)			(1.61)
Illiterate							1.41		2.10
(% population > 14)							(1.55)		(1.71)
Works in primary sector								−0.91	−0.15
(% employed)								(1.53)	(1.56)

Notes: N = 21,571 and $R^2 = 0.36$. All regressions include municipality and time fixed effects, and municipality controls. Standard errors are in parentheses and are clustered at the municipality level. Pre-interventions characteristic are from 1995 except dirt floor and population working in the primary sector, which are from 1990. The third tercile for the variable number of households represents the smallest municipalities.

successful in poorer and/or more rural areas, or perhaps less sanitary areas. Why the program was more successful in smaller municipalities is not clear. Perhaps the program was implemented better in smaller municipalities, or there was greater community support to meet the goals of the program in smaller municipalities. This is an area where future analysis may provide useful insights.

6. Conclusions

This paper investigates the effect of the Mexican conditional cash transfer program, Progresa, on the mortality rate for those aged 65 and older. Taking advantage of the rollout of the program at the national level between 1997 and 2000, we show that the program led to a 4% reduction in the average municipality-level mortality rate. Results are similar for males and females. Results by cause of death showed that the program reduced deaths due to communicable diseases (e.g., infectious diseases), nutrition and anemia, as well as for one non-communicable disease, diabetes. This is important given that death due to diabetes is now in the top 10 causes of death in many middle income countries and is one of the leading causes of death in Mexico (WHO, 2012). It is surprising that the program was not effective at reducing deaths due to typical cardiovascular diseases such as heart disease, stroke and hypertension. Cardiovascular diseases are the leading cause of death in Mexico. However, quality of health care and high out-of-pocket costs to address these diseases may have been an impediment.

There were a number of heterogeneous program effects. The decline in the elderly mortality rate was larger, 6%, in municipalities whose average pre-program mortality rates were above the sample median. However, Progresa also led to significant, though smaller, declines in mortality in those municipalities whose pre-program mortality rate was below the sample median, indicating that even municipalities that may have initially been better off still benefited from the program. In addition, the program effect was larger in smaller municipalities, and in those with less access to electricity and wastewater disposal.

The fact that the program was able to address both communicable and non-communicable diseases is important from a development perspective. Many middle- and low-income countries are in the midst of an epidemiological transition and face the double burden of addressing both communicable and non-communicable diseases. Developing countries also tend to have few programs in place to help the elderly, and these results suggest that modest increases in household income, along with regular preventive check-ups and health education as provided by Progresa, can lead to significant mortality declines. Furthermore, a key lesson for those countries that have a CCT program for which the conditionalities only focus on children, our results highlight that requiring elderly family members to also have regular health care visits may be an effective way to improve the health of these individuals.

Appendix A. Supplementary data

Supplementary data to this article can be found online at http://dx.doi.org/10.1016/j.jdeveco.2013.08.002.

References

Adato, M., Hoddinott, J., 1999. Chapter 22: conditional cash transfer programs: a magic bullet for reducing poverty. In: von Braun, J., Hill, R.V., Pandya-Lorch, R. (Eds.), The Poorest and Hungry: Assessments, Analyses, and Actions. IFPRI, Washington D.C.

Adato, M., Coady, D., Ruel, M., 2000. Final Report: An Operations Evaluation of PROGRESA From the Perspective of Beneficiaries, Promotoras, School Directors, and Health Staff. Washington, DC, International Food Policy Research Institute.

Barham, T., 2011. Providing a healthier start to life: the impact of conditional cash transfer programs on infant mortality. J. Dev. Econ. 94 (1), 74–85.

Bautista-Arredondo, S., S. Bertozzi, P. Gertler, S. Martinez, 2006. How Health Care Supply Constraints Limit the Effectiveness of Conditional Cash Transfer Programs: Evidence from Mexico's Oportunidades Program. Unpublished.

Behrman, J., Parker, S., 2011. The Impact of the PROGRESA/Oportunidades Conditional Cash Transfer Program on Health and Related Outcomes for the Aging in Mexico. PARC Working Paper Series, WPS s11-02.

Bobonis, G. and F. Finan, 2002. Transfers to the poor increase the schooling of the non-poor: the case of Mexico's Progresa Program. Unpublished.

Braine, T., 2006. Mexico's quest for a complete mortality data set. Bull. World Health Organ. 84 (3), 161–256.

Coady, D., 2000. The Application of Social Cost–Benefit Analysis to the Evaluation of Progresa: Final Report. Washington, DC, International Food Policy Research Institute.

Fernald, L., Hou, X., Gertler, P., 2008. Oportunidades program participation and body mass index, blood pressure, and self-reported health in Mexican adults. Prev. Chronic. Dis. 5 (3), 1–12.

Fiszbein, A., Schady, N., 2009. Conditional cash transfers: reducing present and future poverty. World Bank Policy Research Report.The World Bank, Washington D.C.

Frenk, J., González-Pier, E., Gómez-Dantés, O., Lezana, M., Knaul, F., 2006. Heath system reform in Mexico 1: comprehensive reform to improve health system performance in Mexico. Lancet 368, 1524–1534.
Gertler, P., Boyce, S., 2001. An Experiment in Incentive-Based Welfare: The impact of PROGRESA on Health in Mexico. faculty.haas.berkeley.edu/gertler/working_papers/PROGRESA%204-01.pdf.
Gómez-Pérez, F., Rojas, R., Villalpando, S., Barquera, S., Rull, J., Aguilar-Salinas, C., 2010. Prevention of cardiovascular disease based on lipid lowering treatment: a challenge for the Mexican health system. Salud Publica Mex. 52, S54–S62.
Gribble, J., Preston, S., 1993. The Epidemiological Transition: Policy and Planning Implications for Development Counties. National Academy Press, Washington D.C.
Hernández, B., Ramírez-Villalobos, D., Romero, M., Gómez, S., Atkinson, C., Lozano, R., 2011. Assessing quality of medical death certification: concordance between gold standard diagnosis and underlying cause of death in selected Mexican hospitals. Popul. Health Metrics 9 (38), 1–10.
Hoddinott, J., Skoufias, E., 2004. The impact of PROGRESA on food consumption. Econ. Dev. Cult. Chang. 53 (1), 37–61.
Imbens, G., Wooldridge, J., 2009. Recent developments in the econometrics of program evaluation. J. Econ. Lit. 47 (1), 5–86.
Knaul, F., Arreola-Ornelas, H., Méndez, O., January 2005. Catastrophic and Impoverishing Health Expenditure: Increasing Risk Pooling in the Mexican Health System Working Paper Mexican Health Foundation. www.funsalud.org.mx/competitividad/pdf/Risk%20pooling%20en%20Mex%20fin.pdf.
Knaul, F., Arreola-Ornelas, H., Méndez-Carniado, O., et al., 2006. Evidence is good for your health system: policy reform to remedy catastrophic and impoverishing health spending in Mexico. Lancet 368, 1828–1841.
Lozano, R.C., Murray, J. Frenk, Bobadilla, J., 1995. Burden of disease assessment and health system reform: results of a study in Mexico. J. Int. Dev. 7 (3), 555–563.
Mathers, C., Fat, D., Inoue, M., Rao, C., Lopez, A., 2005. Counting the dead and what they died from: an assessment of the global status of cause of death data. Bull. World Health Organ. 83 (3), 171–177.
Mexican Ministry of Health, 2006. Effective Coverage of the Health System in Mexico 2000–2003. D.F., Mexico.
Omran, A., 1971. The epidemiologic transition: a theory of epidemiology of population change. Milbank Mem. Fund Q. 49 (1), 509–538.
Palloni, A., 2002. Living arrangements of older persons. Popul Bull U. N. Spec. Issue 42 (43), 54–110.
Parker, S., Teruel, G., 2005. Randomization and social program evaluation: the case of Progresa. Ann. Am. Acad. Pol. Soc. Sci. 599, 199–219.
Rubalcava, L., Teruel, G., 2006. California Center for Population Research Working Paper 2006-006. papers.ccpr.ucla.edu/papers/PWP-CCPR-2006-006/PWP-CCPR-2006-006.pdf.
Skoufias, E., Davis, B., Vega, S., 1999. Targeting the Poor in Mexico: An Evaluation of the Selection of Households Into OPORTUNIDADES. Washington, DC, International Food Policy Research Institute.
Stecklov, G., Winters, P., Stampini, M., Davis, B., 2005. Do conditional cash transfers influence migration? A study using experimental data from the Mexican PROGRESA program. Demography 42 (4), 769–790.
Stevens, G., Dias, R., Thomas, K., Rivera, J., Carvalho, N., Barquera, S., Hill, K., Ezzati, M., 2008. Characterizing the Epidemiological transition in Mexico: national and subnational burden of diseases, injuries and risk factors. PLoS Med. 5 (6), 900–910.
World Health Organization, 2011. Fact Sheet Number 371: Cardiovascular Diseases [on-line]. [Accessed on June 15 2013] www.who.int/mediacentre/factsheets/fs317/en/index.html.
World Health Organization, 2012. Fact Sheet Number 310: Top Ten Causes of Death [on-line]. [Accessed on June 15 2013] www.who.int/mediacentre/factsheets/fs310.pdf.